

# FAULT HUNT

## Official Event Rulebook

|                    |                                        |
|--------------------|----------------------------------------|
| <b>Prize Pool</b>  | ■10,000                                |
| <b>Team Size</b>   | 1–3 members                            |
| <b>Eligibility</b> | Open to participants from all branches |

## 1. Event Overview

Fault Hunt is a circuit-analysis and troubleshooting-based technical event designed to test participants' understanding of electronic circuits, simulation, debugging, and analytical reasoning. Participants will identify faults in circuits and justify their findings through systematic analysis.

## 2. Event Structure

- The event consists of two rounds conducted on the same day.
- Round 1 is a Preliminary Quiz.
- Round 2 is Circuit Fault Analysis and is conducted for shortlisted teams.

## 3. Round 1 — Preliminary Quiz

- Duration: 30 minutes.
- The quiz covers fundamental concepts of electronics and circuit analysis.
- Performance in this round is used to shortlist teams for the next round.

## 4. Round 2 — Circuit Fault Analysis

- Duration: 1.5 hours.
- Shortlisted teams are provided with LTspice circuit files containing intentional faults.
- Teams analyze the circuits, identify faulty components or connections, and determine the underlying cause of the observed malfunction.
- Participants are evaluated on accuracy of fault identification, circuit analysis and troubleshooting methodology, use of LTspice simulation and relevant measurements, understanding of circuit behavior, thoroughness and clarity of explanation, and logical approach to solving the problem.

## 5. Evaluation

The emphasis is not only on identifying the correct fault, but also on how effectively the team arrives at the solution and explains its reasoning.

## Expected Participation

70 students in Round 1